

END TERM EXAMINATION [MAY-JUNE 2016] FOURTH SEMESTER [B.TECH] ELECTRICAL AND ELECTRONIC MEASURING INSTRUMENTS [ETEE-208]

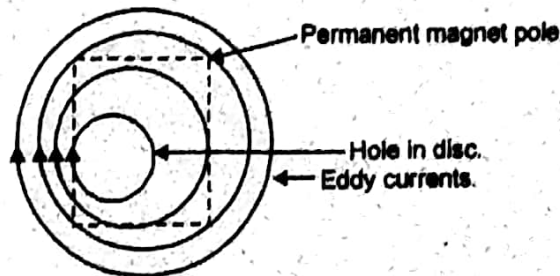
Time : 3 Hrs.

M.M. : 75

Note: Attempt any five questions including Q no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q.1. (a) What is creeping? Explain how it can be prevented. (5)

Ans. In some meters, a slow but continuous rotation is obtained even when there is no current flowing through the current coil and only pressure coil is energized. This is called creeping. The major cause for creeping is over. Compensation for friction because the friction compensating torque is independent of load current as the compensating device is voltage actuated. The other causes are excessive voltage across potential coil, vibrations and stray magnetic field.

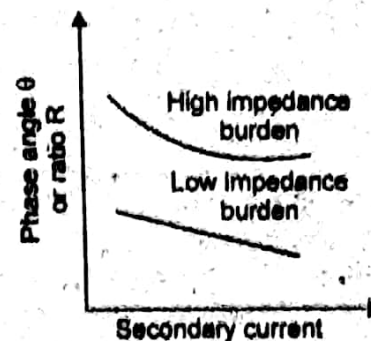


In order to prevent creeping, two diametrically opposite holes are drilled in the disc. The disc will come to rest with one of the holes under the edge of a pole of the shunt magnet. The rotation being thus limited to a maximum of half a revolution.

In some cases, a small piece of iron is attached to the edge of the disc. The force of attraction exerted by the brake magnet on the iron piece is sufficient to prevent creeping of the disc.

Q.1. (b) Explain the effect of change in frequency on the performance of current transformer. (5)

Ans. The effect of increase in frequency will result in proportionate decrease in flux density. Thus, in general, the effect of increase in frequency is similar to that produced by decrease in impedance of secondary winding burden.



Q.1. (c) What are the different types of detectors used for AC Bridges? Mention their frequency range of operation. (5)

Ans. The detectors commonly used for ac bridges are

(i) Head phones, (ii) Vibration Galvanometers, (iii) Tuneable amplifier detectors.

(i) **Head phones:** are widely used as detectors at frequencies of 250 Hz and over 3 or 4 KHz. They are most sensitive detectors for this frequency range.

(ii) **Vibration Galvanometers:** are extremely useful for power and low audio frequency range (5Hz - 1000 Hz and are most commonly used below 200 Hz).

(iii) **Tuneable Amplifier Detectors:** are most versatile of the detectors and works in the frequency range of 10Hz to 100 kHz.

Q.1. (d) Find the frequency of the vertical plates if the frequency applied to the horizontal plate is 75 Hz for the pattern shown below.



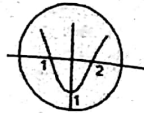
Ans.

$$\frac{f_y}{f_x} = \frac{1}{2}$$

$$\text{Vertical plate frequency} = \text{Total freq.} \times \frac{f_y}{f_x} = 7.5 \times \frac{1}{2} = 37.5 \text{ Hz}$$

$$\frac{f_y}{f_x} = \frac{2}{1}$$

$$\text{Vertical plate frequency} = 75 \times \frac{2}{1} = 150 \text{ Hz}$$



Q.1. (e) What are the advantages of LCD display over Nixie tube and LED display?

Ans. (i) Produces very high brightness image due to high peak intensity. Very suitable for environments that are brightly lit.

(ii) Produce considerably lower electric magnetic and, electromagnetic fields than CRT.

(iii) Consumes less than 1/3rd the power of a comparable CRT.

(iv) Life span 50000 - 100000 hours

(v) Screen size 13-57 inches

(vi) Power consumption: lesser than plasma TV and greater than LED

(vii) Viewing Angle: up to 165°.

UNIT-I

Q.2. (a) A potential transformer, ratio 1000/100 volts has primary resistance = 94.5 Ω, secondary resistance = 0.86 Ω, primary reactance = 66.2 Ω, total equivalent reactance = 110 Ω, no load current = 0.2A at 0.4 power factor. Find phase angle error at no load, burden in VA at unity power factor at which the phase angle will be zero. (6.5)

Ans.

$$\cos \alpha = 0.4$$

$$\sin \alpha = \sqrt{1 - 0.4^2} = 0.917$$

$$I_p = I_0 \cos \alpha = 0.2 \times 0.4 = 0.08 \text{ A}$$

$$I_m = I_0 \sin \alpha = 0.2 \times 0.917 = 0.1834 \text{ A}$$

$$n = \frac{1000}{100} = 10$$

$$0 = \frac{I_p}{n} [(X_p \cos \Delta - R_p \sin \Delta)] + [I_m X_p - I_p R_p]$$

Ans.

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At

$$I_s = 0$$

$$\theta = \frac{I_e X_p - I_m R_p}{n V_s} = \frac{0.08 \times 66.2 - 0.1834 \times 94.5}{10 \times 1000}$$

$$= -41$$

At unity power factor $\cos \Delta = 1$, $\sin \Delta = 0$

$$\theta = \frac{\frac{I_s}{n} \times p + I_e \times p - I_m r_p}{n V_s}$$

For

$$\theta = 0$$

$$\frac{I_s}{n} \times p = I_m r_p - I_e X_p$$

$$I_s = \frac{n}{X_p} [I_m r_p - I_e X_p]$$

$$= \frac{10}{1000} (0.1834 \times 94.5 - 0.08 \times 66.2)$$

$$= 1.09 \text{ A}$$

Burden

$$V_s I_s = 100 \times 1.09 = 109 \text{ VA}$$

Q.2. (b) Describe the Silsbee's method for testing of a current transformer. (6)

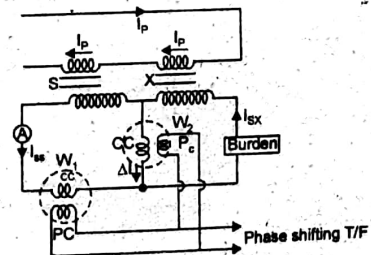
Ans. Silsbee's Method: is a comparing method

Two types

Deflectional

Null

Deflectional Method:



ΔI = Difference between secondary currents of the standard and test transformer.

(i) The phase of the voltage is so adjusted that watt meter W_1 reads zero. Under these conditions, voltage V is in quadrature with current I_s .

$$W_{1q} = V I_s \cos 90^\circ = 0$$

$$W_{2q} = V I_s \sin (\theta_s - \theta_p)$$

$$V = V_s = V_p$$

$$W_{2q} = V I_s \sin (\theta_s - \theta_p)$$

If

$$\begin{aligned}
 W_{1p} &= VI_{1a} \\
 W_{2p} &= V[I_{1a} - I_{1a} \cos(\theta_s - \theta_p)] \\
 &= VI_{1a} - VI_{1a} \cos(\theta_s - \theta_p) \\
 W_{1p} &= VI_{1a} \\
 W_{2p} &= V[I_{1a} \times I_{1a} \cos(\theta_s - \theta_p)] \\
 &= VI_{1a} - VI_{1a} \cos(\theta_s - \theta_p) \\
 &= W_{1p} - VI_{1a} \cos(\theta_s - \theta_p) \\
 &= W_{1p} - VI_{1a} \\
 VI_{1a} &= W_{1p} - W_{2p} \\
 R_x &= I_p / I_{1a} \\
 R_p &= I_p / I_{1a}
 \end{aligned}$$

Phase angle of test transformer

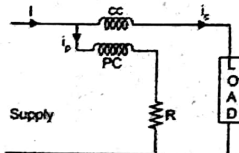
$$\begin{aligned}
 \theta &= \frac{W_{2p}}{W_{1p} - W_{2p}} \neq 0 \\
 &= \frac{W_{2p}}{W_{1p}} + \theta_s \text{ rad.}
 \end{aligned}$$

W_{2p} is very small.

Q.3. (a) Describe the constructional details of an electrodynamicometer type wattmeter. Derive the expression for torque when the instrument is used on a.c. Why it is necessary to make the potential coil purely resistive. (6.5)

Ans. Construction: An electrodynamicometer consists of the following parts:

(i) Fixed Coil: Carry the current of the circuit and are divided into two halves. They are wound with heavy wires and are designed to carry 20A of current.



(ii) Moving Coil: Is mounted on a pivoted spindle and is entirely embraced by the fixed current coils. Spring control is used for the movement. The use of moving coil as pressure coil is a natural consequence of design requirements.

(iii) Control: Spring control is used for instrument.

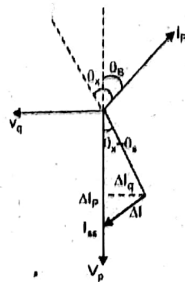
(iv) Damping: Air friction damping is used.

(v) Scales & Pointers: They are equipped with mirror type scales and knife edge pointers to remove reading errors.

$$T_i = k_i \frac{dM}{d\theta}$$

where

T_i = instantaneous torque



i_1, i_2 = instantaneous value of current in two coils

$$v = \sqrt{2} V \sin \omega t$$

$$i_p = V/R_p = \frac{\sqrt{2}V}{R_p} \sin \omega t = \sqrt{2} I_p \sin \omega t$$

$$I_p = V_p/R_p$$

If current lags voltage by angle ϕ

$$i_p = \sqrt{2} I_p \sin(\omega t - \phi)$$

$$T_i = 2 I_p I \sin \omega t \sin(\omega t - \phi) \frac{dM}{d\theta} = I_p I (\cos \phi - \cos(2\omega t - \phi)) \frac{dM}{d\theta}$$

Average deflecting torque

$$T_d = \frac{1}{T} \int_0^T T_i d(\omega t)$$

$$T_d = \frac{1}{T} \int_0^T I_p I (\cos \phi - \cos(2\omega t - \phi)) \frac{dM}{d\theta} d(\omega t) = I_p I \cos \phi \frac{dM}{d\theta} = \frac{VI}{R_p} \cos \phi \frac{dM}{d\theta}$$

Controlling torque $T_c = K\theta$

K = spring constant

θ = final steady deflection

In case inductance is present in the pressure coil,

Actual wattmeter reading will be

$$= \frac{I_p I \cos(\phi - \beta) \frac{dM}{d\theta}}{k}$$

where

$$\beta = \tan^{-1} \left(\frac{\omega L_p}{R_p} \right)$$

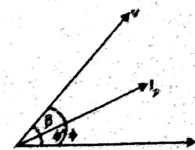
It is clear from the expression that in the absence of inductance, wattmeter reads correctly at all power factors & frequencies.

Q.3. (b) Discuss the various sources of errors in electrodynamicometer type wattmeter. (6)

Ans. Errors in Electrodynamicometer wattmeter:

- (1) Pressure coil inductance
- (2) Pressure coil capacitance
- (3) Errors due to mutual inductance effects
- (4) Errors caused because of connections
- (5) Eddy current errors
- (6) Stray magnetic field errors.
- (7) Errors caused by vibrations of moving systems
- (8) Temperature errors.

1. Pressure coil inductance: Discussed in Last part of Q.3. (a)



2. **Pressure coil capacitance:** Capacitance is mainly due to interturn capacitance of series resistance. The effect is opposite to that of inductance, therefore, the wattmeter reads high on lagging power factors of the load.

3. **Errors due to mutual inductance effect:** (between CC and PC)
These are quite low at power frequencies but they increase and become more important as the frequency is increased.

(4) **Errors caused by vibration of moving system:** This error is found occasionally on alternating currents. The torque on moving systems varies cyclically with a frequency which is twice that of the voltage.

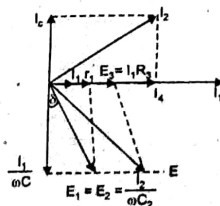
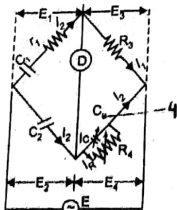
(5) **Temperature errors:** The indication of a wattmeter is effected by changes in room temperature. This is because any change in room temperature changes the resistance of the pressure coil and the stiffness of springs.

UNIT-II

Q.4. (a) Describe the working of low voltage Schering Bridge. Derive the equation for capacitance and dissipation factor. Draw the phasor diagram of the bridge under balance condition. (6.5)

Ans.

- C_1 = capacitor whose capacitance is to be determined.
- r_1 = a series resistance representing loss in C_1
- C_2 = a standard capacitor
- R_2 = non-inductive resistance
- C_4 = variable capacitor
- R_4 = variable non inductive resistance.



$$\text{At balance } \left(r_1 + \frac{1}{j\omega C_1} \right) \left(\frac{R_4}{1 + j\omega R_4 C_4} \right) = \frac{1}{j\omega C_2}$$

$$\left(r_1 + \frac{1}{j\omega C_1} \right) R_4 = \frac{R_4}{j\omega C_2} (1 + j\omega R_4 C_4)$$

$$= r_1 R_4 - \frac{jR_4}{\omega C_1} = \frac{-jR_4}{\omega C_2} \times \frac{R_4 R_4 C_4}{C_4}$$

$$r_1 = R_4 C_4 C_2$$

$$C_1 = C_2 (R_4 / R_2)$$

$$\text{Dissipation factor } D_1 = \tan \delta$$

$$= \omega C_1 r_1 = \omega \left(\frac{C_2 R_4}{R_2} \right) \left(\frac{R_4 C_4}{C_2} \right) = \omega C_4 R_4$$

Q.4. (b) The four arms of a bridge network are made up as follows: 'ab', a resistor of 50Ω in parallel with an inductor of $0.2H$; 'bc' a resistor of 100Ω ; 'cd' and unknown resistor R in parallel with an unknown capacitor C ; 'da', a resistor of 1000Ω . A 50 Hz supply is applied across arm ac. Find R and C when the bridge is under balance condition. (6)

Ans.

$$\left(\frac{1}{R_1} + \frac{1}{j\omega L_1} \right)^{-1} \left(\frac{1}{R_4} + j\omega C_4 \right)^{-1} = R_2 R_3$$

$$\frac{j\omega R_1 L_1}{R_1 + j\omega L_1} \cdot \frac{R_4}{1 + j\omega R_4 C_4} = R_2 R_3$$

$$\frac{j\omega R_1 R_4 L_1}{R_1 + j\omega R_1 R_4 C_4 + j\omega L_1 - \omega^2 L_1 R_4 C_4} = R_2 R_3$$

$$j\omega R_1 R_4 L_1 = R_2 R_3 R_3 - \omega^2 L_1 R_2 R_3 C_4 + j\omega (R_1 R_2 R_3 R_4 C_4 + L_1 R_2 R_3)$$

$$= R_1 R_4 L_1 = R_2 R_3 R_3 + L_1 R_2 R_3$$

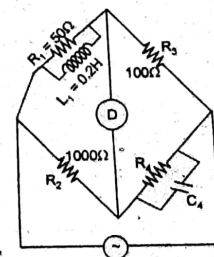
$$R_1 R_2 R_3 = \omega^2 L_1 R_2 R_3 R_4 C_4$$

$$50 = \left(2 \times \frac{22}{7} \times 50 \right)^2 \times 0.2 \times R_4 C_4$$

$$R_4 C_4 = \frac{7 \times 7}{4 \times 22 \times 22 \times 10} = 50 \times 1000 \times 100 \times R_4 C_4 + 0.2 \times 1000 \times 100$$

$$R_4 = \frac{50 \times 1000 \times 100 \times 7 \times 7}{4 \times 22 \times 22 \times 10 \times 10} + \frac{20000}{10} = 3265.5 \Omega$$

$$C_4 = \frac{7 \times 7}{4 \times 22 \times 22 \times 10 \times 3265.5} = 0.775 \mu F$$



Q.5. (a) Explain the function and working of Wagner Earth Devices. (6.5)

Ans. **Wagner Earthing Device:** If each component in a bridge has a defining screen connected to one end, a very high accuracy in measurement is made possible by the addition of a Wagner Earthing Device. It removes all the earth capacitances from bridge network.

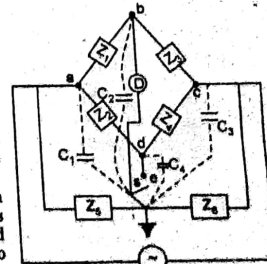
Z_1, Z_2, Z_3, Z_4 = Bridge arms

Z_0, Z_0 = Variable impedance (must be capable of forming a balanced bridge).

If switch S is on d , balance of bridge may be obtained by adjustment of the impedances Z_2 & Z_4 . The presence of earth capacitances will prevent a true balance being obtained, but a point of minimum sound can be obtained.

After adjusting to minimum sound, S is thrown to e . Z_2 & Z_4 are adjusted until minimum sound is obtained. The head phones are now connected to b, d and Z_2 and Z_4 are adjusted to give min sound and so on.

C_1 & C_2 shunting, Z_0 & Z_0 complicate these arms, because of this combination may not give a true balance against Z_2 & Z_4 , only a minimum being attainable during a wagner balance.



Disadvantages can be overcome by using double ratio a.c. bridges.

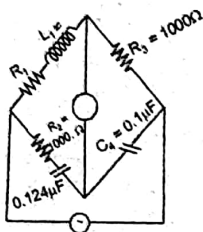
Q.5. (b) An Owen's bridge is used to measure the properties of a sample steel sheet at 2KHz. At balance arm ab is test specimen; arm bc is a resistor of 1K Ω ; arm cd is a capacitor of 0.1 μ F and arm da is a resistor of 1K Ω in series with a capacitor 0.124 μ F. Derive balance condition and calculate the impedance of the specimen under test condition. (6)

Ans.

At balance
$$(R_1 + j\omega L_1) \frac{1}{j\omega C_4} = R_3 \left(R_2 + \frac{1}{j\omega C_2} \right)$$

$$L_1 = R_2 R_3 C_4 = 1000 \times 1000 \times 0.1 \times 10^{-6} \text{ H}$$

$$= 100 \text{ mH} = 0.1 \text{ H}$$



$$R_1 = R_3 \frac{C_4}{C_2} = 1000 \times \frac{0.1}{0.124} = 806.45 \Omega$$

Reactance of specimen at 2 kHz

$$X_L = 2\pi \times 2 \times 1000 \times 0.1 \times 10^{-3} = 125.66 \Omega$$

$$\text{Impedance of specimen } Z_1 = \sqrt{(806.45)^2 + (125.66)^2} = 816.18 \Omega$$

UNIT-III

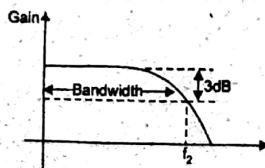
Q.6. (a) Describe in detail the role and working of vertical amplifier used in a CRO. (6.5)

Ans. Vertical amplifier is the principal factor in determining the sensitivity and bandwidth of an oscilloscope. In general, greater sensitivity, which is expressed in terms of V/cm of vertical deflection at midband frequency, is obtained at the expense of bandwidth. As a rule of thumb, an amplifier at some specific cost may be obtained with a certain gain, bandwidth product, which is the product of voltage gain of the amplifier and its bandwidth. Voltage gain may be sacrificed in favour of greater bandwidth, or vice-versa, without significantly affecting the cost of amplifiers.

The sensitivity of an oscilloscope is directly proportional to gain of the vertical amplifier.

Vertical sensitivity is a measure of how much the electron beam will be deflected for a specified input signal. CRT screen is covered with a plastic grid pattern called a graticule. The vertical sensitivity of an oscilloscope is the smallest deflection factor that can be selected with the rotary switch.

The bandwidth of an oscilloscope determines the range of frequencies that can be accurately reproduced on the CRT screen. The greater bandwidth, the wider is the range of frequencies that can be observed with the instrument

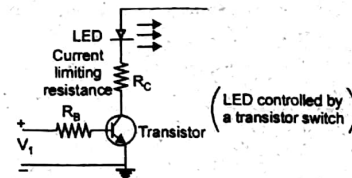
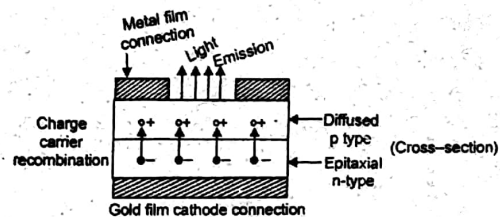


$$t_r \times BW = 0.35$$

t_r = rise time measured in seconds
BW = Bandwidth in Hz.

Q.6. (b) Explain the theory and working of an LED. Describe its advantage. (6)

Ans. LED is a PN junction diode which emits light when a current passes through it in the forward direction.



Charge carrier recombination occurs at a PN junction as electrons cross from N side and recombine with holes on P-side. When recombination takes place, the charge carriers give up energy in the form of heat and light. If the semiconducting material is translucent the light is emitted and the junction is source of light.

Semiconductor material used GaAs produces green light, GaAsP produces red or yellow light.

Advantages of LED

- LEDs are miniature in size
- Output depends upon current flowing through it.
- High efficiency.
- Different colored outputs.
- LED's are rugged and therefore can withstand shocks and vibrations.

Q.7. Explain the following terms applied to digital displays: (12.5)

- Resolution
- Difference between $3\frac{1}{2}$ digit and 4 digit display
- Sensitivity of digital meter
- Accuracy specification of digital meters.

Ans. (a) Resolution: This is sometimes referred to as readability below which differences can no longer be differentiated. In analog instruments the limit is one part in several hundred. Digital instruments can be made with a resolution. Digital instruments can be made with a resolution of one part in several thousands.

The number of digit positions used in a digital meter determines the resolution. Thus a three digit display on digital meter (voltmeter) (DVM) for 0-1V range will be able to indicate 999 mV with smallest increment or resolution of 1mV.

(b) **Difference between $3\frac{1}{2}$ digit and 4 digit display:** In practice a fourth digit to the left of active (usually capable of indicating either 0 or 1 only is placed) digits. This permits going above 999 to 1999 to give overlap between ranges for convenience. This is called over ranging.

This type of display is known as a $3\frac{1}{2}$ digit display.

If n = no. of full digit, resolution = $1/10^n$

A 4 digit display has a resolution of $1/10^4 = 0.0001$ and it can display up to 9999.

(c) **Sensitivity of digital meters:** Is the smallest change in input which a digital meter is able to detect. Therefore it is the lowest voltage range's full scale value multiplied by the meter's resolution.

$$\text{Sensitivity } S = (f.s.)_{\min} \times R$$

where

$(f.s.)_{\min}$ = lowest full scale value of meter

R = Resolution expressed as decimal

(d) **Accuracy specification for Digital meters:** Accuracy is usually expressed as a percentage of the reading plus a percentage of full scale. Often the percentage of full scale part of the statement is given in terms of digits. When this is done, the digit referred to are the least significant digits (LSD).

UNIT-IV

Q.8. (a) How does the sensitivity of an ac voltmeter compare with the sensitivity of a dc voltmeter?

Ans.

(6.5)

DC Voltmeter	AC Voltmeter
(i) It is a voltmeter which can only measure DC voltage	(i) It is a voltmeter which measures AC voltage only
(ii) Sensitivity of DC voltmeter is sensitivity = $\frac{1}{I_{fd}}$	(ii) Sensitivity = $\frac{R_m + R_s}{E} \Omega/V$ R_m = meter resistance R_s = series resistance E = applied voltage
(iii) High sensitivity	(iii) Comparatively lower sensitivity.

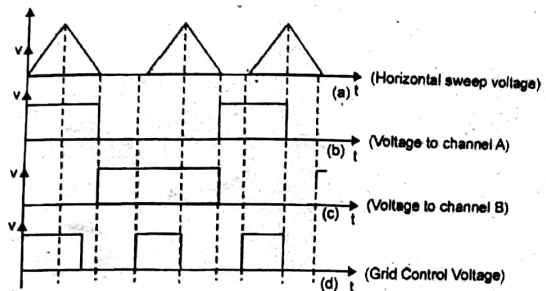
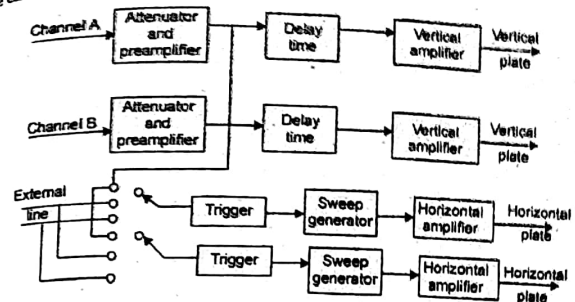
If AC voltage is measured using DC voltmeter, it will read zero value as the average power is zero in that case.

Q.8. (b) Explain the working of Dual beam CRO.

Ans. Dual Beam CRO

Dual trace oscilloscope can not capture two fast transient events, as it can not switch quickly enough between traces. The dual beam oscilloscope has two separate

electron beams, and therefore two completely separate vertical channels. The two channels may have a common time base system or they may have independent time base circuits.



Two methods are used for generating the two electron beams within the CRT

The first method uses a double gun tube. This allows the brightness and focus of each beam to be controlled separately but it is bulkier than a split beam tube.

The second method, known as splitbeam, a single gun is used. It has half brightness of a single beam, which has disadvantages at higher frequency operation.

Q.9. Describe the working of a sweep frequency generator. What are the sweeper errors? (12.5)

Ans. Sweep frequency generator: is a special type of signal generator in which the output frequency is cyclically swept through a range of frequencies. The instrument may have a display built into it, to show the variation of amplitude with frequency, or it may provide signals for use with an external oscilloscope.

Two modes are used to set the swept frequency range

(i) The stop start mode ($f_1 - f_2$)

(ii) Delta frequency mode.

Three approaches may be used to cover a wide range of frequency from a single instrument:

(i) Stacked switching

(iii) Heterodyne control.

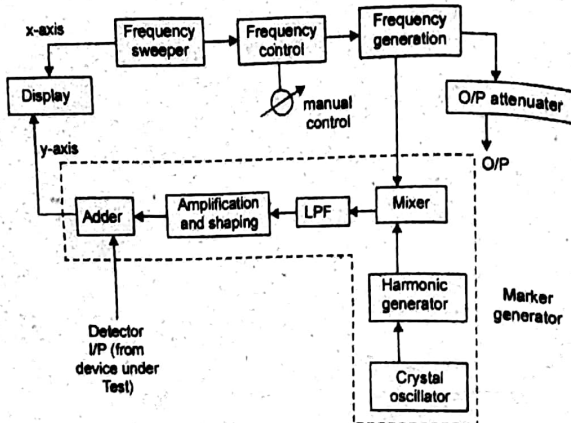
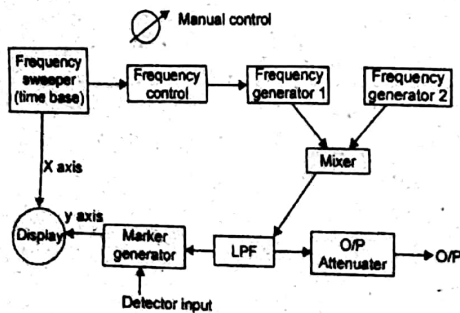


Fig. Block diagram of a sweep generator using a heterodyne frequency generator

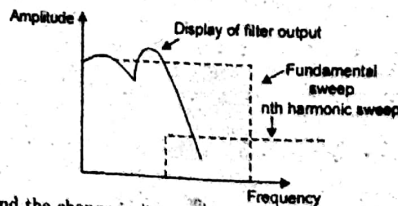


Sweeper Errors: Several errors occur in swept frequency generator measurements, due to imperfection in the frequency sources will be swept along with the fundamentals, but at a faster rate.

Effect of sweep generator harmonics on a LPF response measurement.

The accuracy of the output, and the change in its amplitude with frequency, is another important consideration. The amplitude varies with the sweep rate, since the amplitude feedback correction. Circuit have less time to work at faster sweep rate.

The output from a sweeper is not linear, which means that the spacing between markers is unequal, so that one can not easily estimate a value at a frequency between two markers.



FIRST TERM EXAMINATION [FEB. 2017] FOURTH SEMESTER [B.TECH] ELECTRICAL AND ELECTRONIC MEASURING INSTRUMENTS [ETEE-208]

Time : 1.30 hrs.

M.M. : 80

Note: Attempt any three questions including Q.no. 1 which is compulsory.

Q.1. (a) Discuss major sources of errors in Current transformers. (8)

Ans. Ratio Error: The value of transformation ratio is not equal to the turns ratio. Also, the value is not constant but depends upon the magnetising and loss components of the exciting current, the secondary winding load current and its power factor.

It can be defined as

$$\begin{aligned} \% \text{ Ratio error} &= \frac{\text{Nominal ratio} - \text{Actual ratio}}{\text{Actual ratio}} \times 100 \\ &= \frac{K_n - R}{R} \times 100 \end{aligned}$$

This causes considerable error in the current measurement correction, Taking δ small.

$$R = n + \frac{I_p}{I_s}$$

$$= n \left(1 + \frac{I_p}{I_s} \right)$$

$$\begin{aligned} \sin \delta &= 0 \\ \cos \delta &= 1 \\ n &= \frac{I_p}{I_s} \end{aligned}$$

Phase Angle Error: In power measurement, it is necessary that phase of secondary winding current shall be displaced by exactly 180° from that of primary winding current. It is seen that phase difference differs by an angle θ from 180° .

Hence, this introduces considerable error into current transformer can be defined as,

$$\theta = \frac{180}{\pi} \left[\frac{I_m \cos \delta - I_p \sin \delta}{n I_s} \right] \text{ degree}$$

At $\delta \rightarrow$ small & positive.

$$\theta = \frac{180}{\pi} \left[\frac{I_m}{n I_s} \right] \text{ degree}$$

But, $I_p = n I_s$

$$\theta = \frac{180}{\pi} \left[\frac{I_m}{I_p} \right] \text{ degree}$$

Q.1. (b) Classify resistances and name instruments to measure resistance of different values. (2)

Ans. Pg-4-2016, Q.4(b) First Term Examination, feb-2016.

Q.1. (c) What are the different detectors used in A.C Bridges? (2)

Ans. Pg-2-2016, Q.1(c) First Term Examination, feb-2016.

Q.1. (d) Two wattmeters are connected to measure power of three phase circuit with power factor of 0.35. Total power consumed by the load, as indicated by two wattmeters is 70 kW. Find the individual wattmeter reading. (3)

Ans.

$$\begin{aligned} \text{p.f.} &= 0.35 \\ P_1 + P_2 &= 70 \text{ kW} \\ \cos \phi &= 0.35 \\ \phi &= \cos^{-1} 0.35 = 69.51^\circ \end{aligned}$$

...(1)

$$\tan \phi = \sqrt{3} \left(\frac{P_1 - P_2}{P_1 + P_2} \right)$$

$$\tan 69.51^\circ = \sqrt{3} \left(\frac{P_1 - P_2}{70 \times 10^3} \right)$$

$$2.67 \times 70 \times 10^3 = \sqrt{3} (P_1 - P_2)$$

$$P_1 - P_2 = 107.9 \text{ kW} \quad \dots(2)$$

Adding eq. (1) and (2), we get

$$2P_1 = 177.9$$

$$P_1 = 88.95 \text{ kW}$$

Subtracting eq. (2) from eq. (1), we get

$$2P_2 = -37.9$$

$$P_2 = -18.95 \text{ kW}$$

Q.2. (a) An electrodynamicometer type wattmeter is used for power measurement of a load at 100 V and 9 A at a power factor of 0.1 lagging. The pressure coil current has resistance of 3000 Ω and inductance of 30 mH. Calculate the % error in wattmeter when the pressure coil is connected on the load side. Supply frequency is 50 Hz. (5)

Ans. Power consumed by the load = $VI \cos \phi$

$$= 100 \times 9 \times 0.1$$

$$= 90 \text{ W}$$

The wattmeter measures the loss in the pressure coil circuit for this connection.

$$\text{Loss in the pressure coil circuit} = \frac{V^2}{R_p} = \frac{(100)^2}{3000}$$

$$= 3.33 \text{ W}$$

$$\% \text{ Error} = \frac{3.33}{90} \times 100 = 3.7 \%$$

Q.2. (b) Explain in detail construction and working of electrodynamicometer type single phase power factor meter. (5)

Ans. The construction of a single-phase electrodynamicometer type power factor meter is shown in Fig. 1. It consists of a fixed coil which acts as the current coil. This coil is split up into two parts and carries the current of the circuit under test. Therefore, the magnetic field produced by this coil is proportional to the main current.

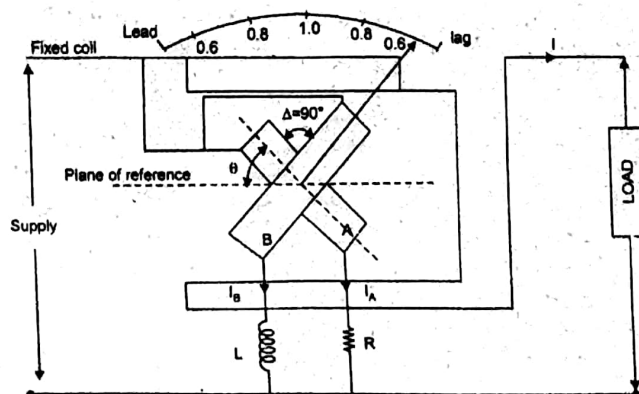


Fig. Single-Phase electrodynamicometer Type power factor meter.

Two identical pressure coils A & B pivoted on a spindle constitute the moving system. Pressure coil A has a non-inductive resistance R connected in series with it, and coil B has a highly inductive choke coil L connected in series with it. The two coils are connected across the voltage of the circuit. The values of R and L are so adjusted that the two coils carry the same value of current at normal frequency, i.e. $R = \omega L$. The current through coil A is in phase with the circuit voltage while that through coil B lags the voltage by an angle Δ which is nearly equal to 90° . The angle between the planes of coils is also made

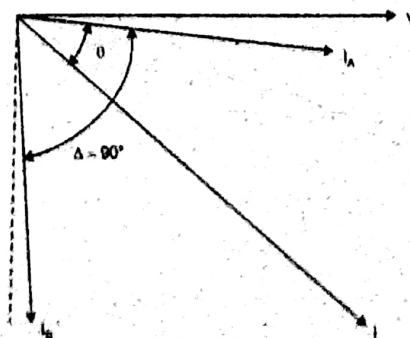


Fig. Phasor diagram for Fig. 1.

equal to Δ . There is no controlling device. Connections to moving coils are made through thin silver or gold ligaments which are extremely flexible and thus give a minimum control effect on the moving system.

In order to simplify the problem we assume that the current through coil B lags the voltage by exactly 90° . Also, that the angle between planes of coils is exactly 90° .

i.e. $\Delta = 90^\circ$.

Now, there will be two deflecting torques, one acting on coil A and the other on coil B. The coil windings are so arranged that the torques due to the two coils are opposite in direction. Therefore, the pointer will take up a position where these two torques are equal.

Thus consider the case of a lagging power factor $\cos \phi$.

Deflecting torque acting on coil A is:

$$T_A = KVI M_{\max} \cos \phi \sin \theta$$

where,

θ = Angular deflection from the plane of reference.

$M_{\max}$ = Maximum value of mutual inductance between the two coils.

This torque say acts in the clockwise direction.

Deflecting Torque acting on coil B is:

$$T_B = KVI M_{\max} \cos(90 - \phi) \sin(90 + \theta) \\ = KVI M_{\max} \sin \phi \cos \theta$$

This torque acts in the anticlockwise direction. The value of $M_{\max}$ is the same in the two expressions owing to similar constructions of the coils.

The coils will take up such a position that the two torque are equal.

Hence at equilibrium, $T_A = T_B$

or $KVI M_{\max} \cos \phi \sin \theta = KVI M_{\max} \sin \phi \cos \theta$

$\Rightarrow$

$$\boxed{\theta = \phi}$$

Therefore, the deflection of the instrument is a measure of phase angle of the circuit. The scale of the instrument can be calibrated in directly in terms of power factor.

Q.3. (a) The four arms of Maxwell's inductance-capacitance Bridge at balance are, branch AB has choke coil L_1 and series resistance R_1 , Branch BC has non-inductive resistance 800Ω , Arm CD has capacitance of $0.3 \mu F$ in parallel with a noninductive resistance of 800Ω , Arm DA has a noninductive resistance of 800Ω . Find the value of L_1 and R_1 at balance condition. Supply is connected between C and D terminal. (6)

Ans. L_1 and R_1 are in series in branch AB.

$$R_3 = 800 \Omega$$

$$C_4 = 0.3 \mu F$$

$$R_4 = 800 \Omega$$

$$R_2 = 800 \Omega$$

Since supply is connected between A and C, at balance

$$(R_1 + j\omega L_1) \left(\frac{R_4}{1 + j\omega C_4 R_4} \right) = R_2 R_3$$

$$R_1 R_4 + j\omega L_1 R_4 = R_2 R_3 + j\omega R_2 R_3 C_4 R_4$$

$$R_1 R_4 = R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4} = \frac{800 \times 800}{800} = 800 \Omega$$

$$L_1 R_4 = R_2 R_3 C_4 R_4$$

$$L_1 = \frac{R_2 R_3 C_4}{R_4} \\ = \frac{800 \times 800 \times 0.3 \times 10^{-6}}{800} \\ = 192 \text{ mH}$$

Q.3. (b) Derive balance condition for Kelvin Double bridge, explain how resistance of wire has no effect on the measurement. (4)

Ans. Kelvin Double Bridge Method of Measurement of Low Resistances: The kelvin bridge is modification of the wheatstone bridge and provides greatly increased

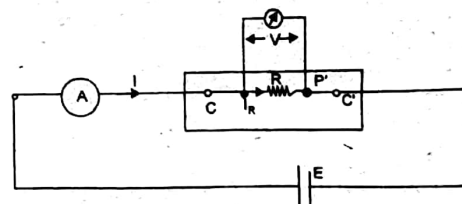


Fig. Ammeter-voltmeter method for measuring 4 terminal resistance accuracy in measurement of low value resistances. An understanding of the Kelvin bridges arrangement may be obtained by a study of the difficulties that arise in a Wheatstone bridge on account of the resistance of the leads and the contact resistances while measuring low values resistors.

Consider the bridge circuit shown in Fig. A where r represents the resistance of the lead that connects the unknown resistance R to standard resistance S . Two galvanometer connections indicated by dotted lines, are possible. the connection may be either to

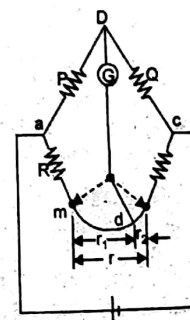


Fig. B Illustrating principle of kelvin's bridge.

point 'm' or to point 'n'. When the galvanometer is connected to point m, the resistance, r , of the connecting leads to the standard resistance, S , resulting in indication of too low an indication for unknown resistance R . When the connection is made to point n, the resistance, r , is added to the unknown resistance resulting in indication of too high a value for R .

Suppose that instead of using point m, which gives a low result, or n, which makes the result high, we make the galvanometer connection to any intermediate point 'd'. If at point 'd' the resistance r is divided into two parts, r_1 and r_2 , such that

$$\frac{r_1}{r_2} = \frac{P}{Q} \quad \dots(1)$$

Then the presence of r_1 , the resistance of connecting leads, causes no error in the result. We have,

$$R + r_1 = \frac{P}{Q} (S + r_2) \text{ but } \frac{r_1}{r_2} = \frac{P}{Q} \quad \dots(2)$$

or

$$\frac{r_1}{r_1 + r_2} = \frac{P}{P+Q} \text{ or } r_1 = \frac{P}{P+Q} \cdot r \text{ as } r_1 + r_2 = r$$

and

$$r_2 = \frac{Q}{P+Q} \cdot r$$

$\therefore$ We can write Eqn. (2) as $\left(R + \frac{P}{P+Q} r\right) = \frac{P}{Q} \left(S + \frac{Q}{P+Q} r\right)$ or $R = \frac{P}{Q} S$. $\dots(3)$

Therefore we conclude that making the galvanometer connection as at c, resistance of leads does not affect the result.

The process described above is obviously not a practical way of achieving the desired result, as there would certainly be a trouble in determining the correct point for galvanometer connections. It does, however, suggest the simple modification, that two actual resistances units of correct ratio be connected between points m and n, the galvanometer be connected to the junction of the resistors. This is the actual Kelvin bridge arrangement, which is shown in Fig. C

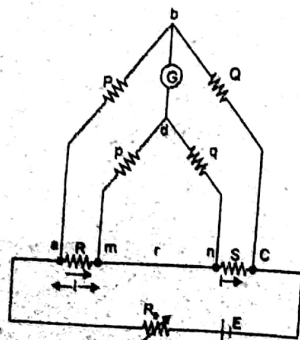


Fig. C. Kelvin double bridge

The Kelvin double bridge incorporates the idea of a second set of ratio arms-hence the name double bridge-and the use of four terminal resistors for the name double bridge-and the use of four terminal resistors for the low resistance arms. Fig C shows the schematic diagram of the Kelvin bridges. The first of ratio arms is P and Q . The second set of ratio arms, p and q is used to connect the galvanometer to a point d at the appropriate potential between points m and n to eliminate the effect of connecting lead of resistance r between the known resistance, R , and the standard resistance, D .

The ratio p/q is made equal to P/Q . Under balance conditions there is no current through the galvanometer, which means that the voltage drop between a and b , E_{ab} is equal to the voltage drop E_{amd} between a and c .

$$\text{Now } E_{ab} = \frac{P}{P+Q} E_{ac} \text{ and } E_{ac} = I \left[R + S + \frac{(p+q)r}{p+q+r} \right] \quad \dots(4)$$

and

$$E_{amd} = I \left[R + \frac{p}{p+q} \left(\frac{(p+q)r}{p+q+r} \right) \right] = I \left[R + \frac{pr}{p+q+r} \right] \quad \dots(5)$$

For zero galvanometer deflection, $E_{ab} = E_{amd}$

$$\text{or } \frac{P}{P+Q} I \left[R + S + \frac{(p+q)r}{p+q+r} \right] = I \left[R + \frac{pr}{p+q+r} \right]$$

or

$$R = \frac{P}{Q} S + \frac{qr}{p+q+r} \left[\frac{P}{Q} - \frac{p}{q} \right] \quad \dots(6)$$

$$\text{Now, if } P/Q = p/q, \text{ Eqn. (6) becomes, } R = \frac{P}{Q} S \quad \dots(7)$$

Eqn. 7 is the usual working equation for the kelvin bridge. It indicates that the resistance of connecting lead, r , has no effect on the measurement, provided that the two sets of ratio arms have equal ratios. Eqn 6 is useful, however, as it shows the error that is introduced in case the ratios are not exactly equal. It indicates that it is desirable to keep r as small as possible in order to minimize the error in case there is a difference between ratios P/Q and p/q .

The effect to thermo-electric emfs can be eliminated by making another measurement with the battery connections reversed. The true value of R being the mean of the two readings.

In a typical kelvin bridges, the range of resistance covered is $0.1 \mu\Omega$ to 1.0Ω . The accuracies are under:

From $1000 \mu\Omega$ to 1.0Ω : 0.05 %

From $100 \mu\Omega$ to $1000 \mu\Omega$: 0.2% to 0.05%.

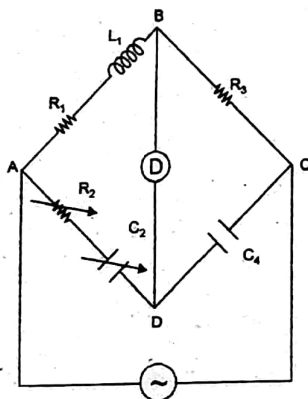
From $10 \mu\Omega$ to $100 \mu\Omega$: 0.5% to 0.2, limited by thermoelectric emfs.

In this bridges there are four internal resistance standards of 1Ω , 0.1Ω , 0.01Ω and 0.001Ω respective.

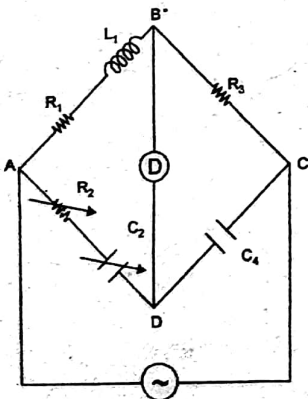
Q.4. (a) Draw a schematic diagram showing constructional details of an induction type energy meter and label its different parts also explain working of induction type energy meter. (6)

Ans. Pg-21-2015. Q.3(a) End Term Examination [May-June-2015.]

Q.4. (b) Derive balance condition of A.C bridge arrangement give in the diagram, also draw phasor diagram of the circuit. (4)



Ans.



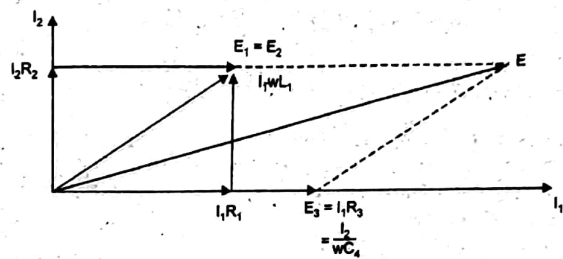
At balance,

$$(R_1 + j\omega L_1) \left(\frac{1}{j\omega C_2} \right) = \left(R_2 + \frac{1}{j\omega C_4} \right) R_3$$

Separating the real and imaginary terms, we get

$$L_1 = R_2 R_3 C_4$$

$$R_1 = R_3 \frac{C_4}{C_2}$$



Phasor diagram

**END TERM EXAMINATION [MAY-JUNE 2017]
FOURTH SEMESTER [B.TECH.]
ELECTRICAL AND ELECTRONIC MEASURING
INSTRUMENT [ETEE-208]**

M.M. : 75

Time : 3 hrs.

Note: Attempt any five questions including Q. No. 1 which is compulsory. Select one question from each unit.

Q.1.(a) Describe the errors in electro-dynamometer type wattmeters. (3)

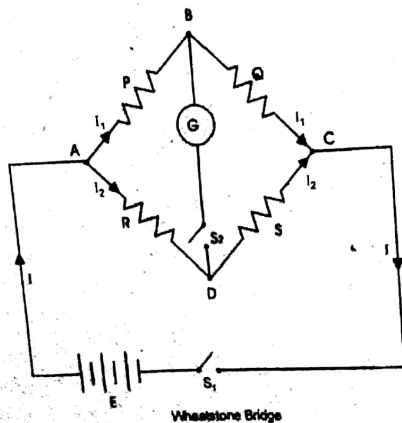
Ans. Following are the errors in the electro-dynamometer type wattmeters:

1. Errors in the pressure coil inductance.
2. Errors may be due to pressure coil capacitance.
3. Errors may be due to mutual inductance effects.
4. Errors may be due connections (i.e. pressure coil is connected after current coil)
5. Error due to Eddy currents.
6. Errors caused by vibration of moving system.
7. Temperature error.
8. Errors due to stray magnetic field.

Q.1. (b) Draw the circuit of a Wheatstone bridge and derive the condition of balance. (3)

Ans. Now from wheatstone bridge circuit

$$\text{current } I_1 = \frac{V}{P+Q}$$



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and

$$\text{current } I_2 = \frac{V}{P+Q}$$

Now potential of point B in respect of point C is nothing but the voltage drop across the resistor Q and this is

$$I_1 Q = \frac{VQ}{P+Q} \dots\dots(i)$$

Again potential of point D in respect of point C is nothing but the voltage drop across the resistor S and this is

$$I_2 S = \frac{VS}{R+S} \dots\dots(ii)$$

Equation, equations (i) and (ii) we get,

$$\begin{aligned} \frac{VQ}{P+Q} &= \frac{VS}{R+S} \Rightarrow \frac{Q}{P+Q} = \frac{S}{R+S} \\ \Rightarrow \frac{P+Q}{Q} &= \frac{R+S}{S} \Rightarrow \frac{P}{Q} + 1 = \frac{R}{S} + 1 \Rightarrow \frac{P}{Q} = \frac{R}{S} \\ &\Rightarrow R = S \times \frac{P}{Q} \end{aligned}$$

Here in the above equation, the value of S and P/Q are known, so value of R can easily be determined.

Q.1. (c) Discuss the advantages of digital indicating instrument over their analog counterparts. (3)

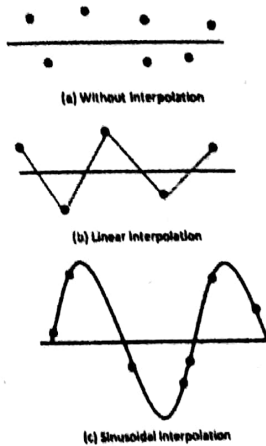
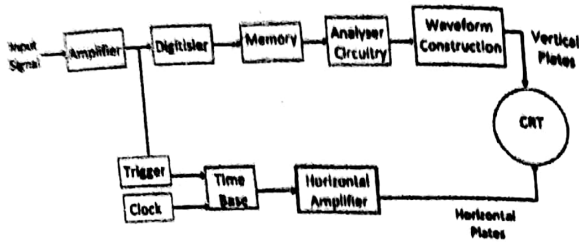
Ans. Advantages of Digital indicating instruments over analog counterparts are -

1. In this limit can be made with a resolution of one part in several thousands.
2. Its accuracy is much higher.
3. Most Digital instruments are D.C instruments that measure up to 100V to 1kV by means of a range attenuator.
4. These instruments are free from observational errors, like a parallax and approximation errors.
5. They are complex & consist of large number of parts which individually react to change in temperature and humidity.

Q.1. (d) Describe the principle of working of digital oscilloscope. (3)

Ans. Working Principle of Digital Storage Oscilloscope

The digital oscilloscope digitises and stores the input signal. This can be done by the use of CRT (Cathode ray tube) and digital memory. The block diagram of the basic digital oscilloscope is shown in the figure below. The digitisation can be done by taking the sample input signals at periodic waveforms.



The maximum frequency of the signal which is measured by the digital oscilloscope depends on the two factors. These factors are the

1. Sampling rate
2. Nature of converter.

Sampling Rate - For safe analysis of input signal the sampling theory is used. The sampling theory states that the sampling rate of the signal must be twice as fast as the highest frequency of the input signal. The sampling rate means analogue to digital converter has a high fast conversion rate.

Converter - The converter uses the expensive flash whose resolution decreases with the increases of a sampling rate. Because of the sampling rate, the bandwidth and resolution of the oscilloscope are limited.

The need of the analogue to digital signal converters can also be overcome by using the shift register.

The input signal is sampled and stored in the shift register. From the shift register, the signal is slowly read out and stored in the digital form. This method reduces the cost of the converter and operates up to 100 mega sample per second.

The only disadvantage of the digital oscilloscope is that it does not accept the data during digitisation, so it had a blind spot at that time.

Waveform Reconstruction

For visualising the final wave, the oscilloscopes use the technique of interpolation. The interpolation is the process of creating the new data points with the help of known variable data points. Linear interpolation and sinusoidal interpolation are the two processes of connecting the points together.

In interpolation, the lines are used for connecting the dot together. Linear interpolation is also used for creating the pulsed or square waveform. For sine waveform, the sinusoidal interpolation is utilised in the oscilloscope.

Q.1. (a) Describe the methods to measure the voltage and power at radio frequencies.

Ans. Techniques for calibration an rf voltmeter

In the absence of a standard rf voltage measuring instrument for a sine wave signal, the voltage at a reference plane in a coaxial transmission line may be calculated from a knowledge of the incident rf power that is the power which would be absorbed in perfectly terminated line, and the rf impedance at the reference plane in accordance with the expression

$$V = (P \times Z)^{1/2}$$

The two calibration methods which are widely used to meet present rf voltage calibration requirements are:

(a) A comparator reference plane voltage calibration

By referring all rf voltage measurements to the central reference plane of a T-junction (Figure 1), there is no longer a need to measure the impedance of the unknown voltmeter. As far as the standard instrument is concerned, provided it presents a good match to the transmission line system it can be considered that the modulus of the voltage reflection coefficient may be of any phase at frequencies where electrical path lengths are possibly significant. The upper frequency for insignificance in path length depends on VRC magnitude.

For the Rohde and Schwarz NRS power sensor used in the laboratory, the frequency will be taken as 10MHz.

(b) An incident voltage calibration

A calibration in terms of the rf voltage which would be developed in a perfectly matched 50 ohm coaxial transmission line system is referred to as an incident voltage

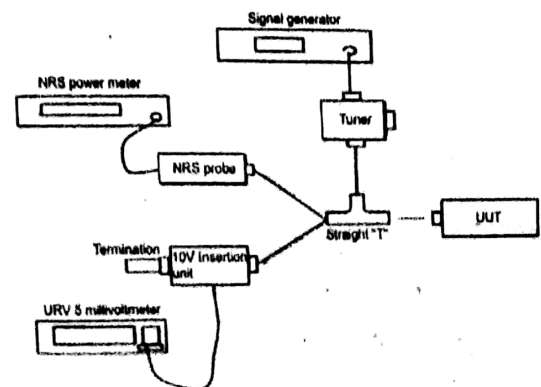
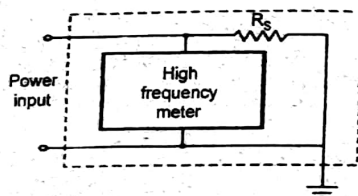


Fig. 1. Comparator configuration.

calibration. It is required for the calibration of the output of signal generators or for the calibration of voltmeters in terms of incident voltage. Although the impedance Z in the expression for voltage may be taken as 50 ohms, deviations of source impedance or voltmeter impedance from 50 Ohms may contribute to uncertainty due to voltage wave reflections.

Power Measurement at radio frequency:

The method of using a dummy load and a voltmeter can be used at radio frequencies upto 500 MHz also. In such a case, the voltmeter and load resistor are usually combined into a single absorption type power meter as shown in the Fig.



The resistance R_s is designed that its value remains constant over the entire range of frequencies of interest. The voltmeter is a high frequency meter which is capable of responding to high frequency signals with high accuracy. The power is calculated by using the basic relationship as.

$$P = \frac{V_s^2}{R_s}$$

where

V_s = voltmeter reading

In practice the standard resistors are available which have constant value for frequencies ranging from d.c. to 4 GHz. This type of power meter is used to measure power below 500 MHz only.

To measure power at high frequency from 500 MHz to 40 GHz two special type of absorption meters are popularly used. These meters are

1. Calorimeter power meter
2. Bolometer power meter

Both these meters use the sensing of heating effects caused by the power signal to be measured

UNIT-I

Q.2. (a) Describe how high current and voltage are measured with the help of instrument transformer. Draw the diagram and discuss the advantages of instrument transformer. (8)

Ans. The extension of instrument range, so that current, voltage power and energy can be measured with instruments or meters of moderate size is of very great importance in measurements are not possible as these currents and voltages are very large and therefore, direct size and cost. The solution lies in stepping down these currents and voltages with the help of instrument transformers so that they could be metered with instruments of moderate sizes.

Fig. 1. shown current being measured by a C.T. The primary winding is so connected that the current being measured passes through it and the secondary winding is connected to an ammeter. The "C.T." steps down the current to the level of ammeter.

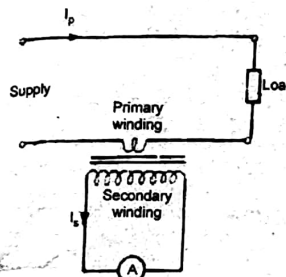


Fig. 1. Current transformer

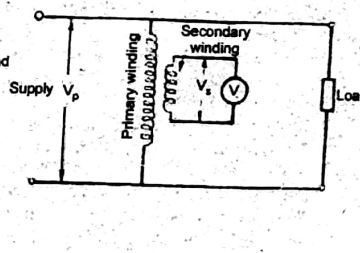


Fig. 2. Potential transformer

Fig. 2. shows voltage measurement with a P.T. The primary winding is connected to the voltage being measured and the secondary winding, to a voltmeter. The 'P.T.' steps down the voltage to the level of voltmeter.

It might appear that extension of range could be conveniently done by the use of shunts for currents and multipliers for voltage measurements, as is done in d.c. measurements. But this method is suitable only for small value of current and voltage. There are many disadvantages with the use of shunts and multipliers.

Advantages of Instrument Transformers

(i) When instruments are used in conjunction with instruments transformers, their readings do not depend upon their constants (R.L.C) as is the case with shunts and multipliers. The instruments transformers produce practically the same instrument reading regardless of the constants of the instruments or, in fact, the number of instruments connected in the circuit.

(ii) Current transformers have been standardized at 5 A secondary winding current and the voltage transformers at from 100 to 120 V secondary winding voltage. These are very moderate ratings and the instruments for measurements are rated near these. Thus a 5 A ammeter may be used to measure 1000 A with the help of a 1000/5 A ratio current transformer of a 110 V voltmeter may be used to measure a voltage of 66 kV with the help of a 66,000/110 V potential transformer. Therefore, very cheap moderate rating instruments may be used to measure large currents and high voltage.

(iii) With the standardization of C.T. and P.T. secondary winding ratings, it is possible to standardize instruments around these ratings and therefore, there is great reduction in the costs of instruments transformers and instruments. Also, replacement of instrument transformers is easy on account of the standardization of the ratings.

(iv) The measuring circuit is isolated from the power circuit. The fact that the meters in the secondary circuit of an instruments transformer are isolated electrically from the primary side is of very great importance in high voltage systems. Large alternators usually operate at voltage between 11 kV to 20 kV while transmission voltage as high as 750 kV are being used. Therefore, in all electrical, systems it is necessary to have a measure of currents and voltages, both for metering purposes and for the operation of protective relays and other equipment. It is impossible to bring the

high voltage lines directly to the switchboard to be connected to instruments as even for voltages upto a few thousand volts it would be difficult to insulate the equipment to provide safety for the operating personnel.

Q.2. (b) Describe the constructional detail of an electro-dynamometer type wattmeter. Derive the expression for the torque when the instrument is used on a.c.

Ans. Electro-dynamometer Wattmeters

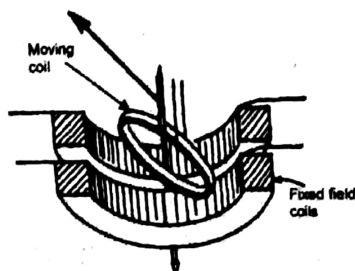
These instruments are similar in design and construction to electro-dynamometer type ammeters and voltmeters.

- The two coils are connected in different circuits for measurement of power.
- The fixed coils or "field coils" are connected in series with the load and so carry the current in the circuit.
- The fixed coils, therefore, form the current coil or simply C.C. of the wattmeter.
- The moving coil is connected across the voltage and, therefore, carries a current proportional to the voltage.
- A high non-inductive resistance is connected in series with the moving coil to limit the current to a small value.
- Since the moving coil carries a current proportional to the voltage, it is called the "pressure coil" or "voltage coil" or simply called. P.C. of the wattmeter.

Construction of Electro-dynamometer Wattmeter

1. Fixed Coils: The fixed coils carry the current of the circuit. They are divided into two halves. The reason for using fixed coils as current coils is that they can be made more massive and can be easily constructed to carry considerable current since they present no problem of leading the current in or out. The fixed coils are wound with heavy wire. This wire is stranded or laminated especially when carrying heavy currents in order to avoid eddy current losses in conductors.

The fixed coils of earlier wattmeters were designed to carry a current of 100A but modern designs usually limit the maximum current ranges of wattmeters to about 20. A for power measurements involving large load currents, it is usually better to use a 5A wattmeter in conjunction with a current transformer of suitable range.



Dynamometer wattmeter

2. Moving Coil: The moving coil is mounted on a pivoted spindle and is entirely embraced by the fixed spindles & is entirely embraced by the fixed current coils. Spring control is used for the movement. The use of moving coil as pressure coil is a natural

consequence of design requirements. Since the current of the moving coil is carried by the instrument springs, it is limited to values, which can be carried safely by springs.

3. Control: Spring control is used for the instrument.

4. Damping: Air friction damping is used. The moving system carries a light aluminium vane which moves in a sector shaped box. Electromagnetic or eddy current damping is not used as introduction of a permanent magnet (for damping purposes) will greatly distort the weak operating magnetic field. Scales and Pointers. They are equipped with mirror type scales and knife edge pointers to remove reading errors due to parallax.

Theory of Electro-dynamometer Watt-meters

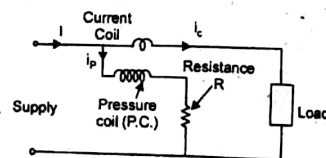
The instantaneous torque of an electro-dynamometer instruments is given by:

$$T_i = i_1 i_2 dM/d\theta$$

where, i_1 and i_2 are instantaneous values of currents in two coils. Let V and I be the rms values of voltage and current being measured.

∴ Instantaneous value of voltage across the pressure coil circuit is

$$v = \sqrt{2}V \sin \omega t$$



Circuit of electro-dynamometer

If the pressure coil circuit has a very high resistance, it can be treated as purely resistive. Therefore, current, i_p in the pressure coil is in phase with the voltage and its instantaneous value is

$$i_p = v/R_p = \sqrt{2}V/(R_p) \sin \omega t = \sqrt{2}I_p \sin \omega t$$

here

$$I_p = V/R_p = \text{rms value of current in pressure coil circuit}$$

and R_p = resistance of pressure coil circuit

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is

$$i_c = \sqrt{2}i \sin(\omega t - \phi)$$

$$\therefore \text{Instantaneous torque } T_i = \sqrt{2}I_p \sin \omega t \times \sqrt{2}I \sin(\omega t - \phi) dM/d\theta$$

$$= 2I_p I \sin \omega t \sin(\omega t - \phi) dM/d\theta = I_p I (\cos \phi - \cos(2\omega t - \phi)) dM/d\theta$$

It is clear from above that there is a component of power which varies as twice the frequency of current and voltage (mark the term containing $2\omega t$).

Average deflecting torque

$$T_d = \frac{1}{T} \int_0^T T_i d(\omega t) = \frac{1}{T} \int_0^T I_p I [\cos \phi - \cos(2\omega t - \phi)] \frac{dM}{d\theta} d(\omega t)$$

$$= I_p I \cos \phi dM/d\theta$$

$$= (VI/R_p) \cos \phi \, dM/d\theta$$

Controlling torque exerted by springs $T_c = K\theta$

Where K = spring constant: θ = final steady deflection.

Since, the moving system of the instrument cannot follow the rapid vibrations in torque, it will take up a position at which the average deflection torque is equal to the restoring torque of the springs.

Therefore, at balance position, $K\theta = I_p I \cos \phi \, dM/d\theta$

or deflection,

$$\theta = \frac{I_p I \cos \phi (dM/d\theta)}{K} (VI \cos \phi / R_p K) dM/d\theta$$

$$= K_1 VI \cos \phi \, dM/d\theta = (K_1 dM/d\theta) P$$

P = power being measured = $VI \cos \phi$

where,

$$K_1 = 1/R_p K$$

Q.3. (a) A currently adjusted, single phase 240V induction watt hour meter has a meter constant of 600 revolutions per kWh. Determine the speed of disc, for the current of 10 A at a power factor of 0.8 lagging. If the lag adjustment is altered so that the phase angle between voltage coil flux and applied voltage is 86° . Calculate the error introduced at (i) unity p.f. (ii) 0.5 p.f. lagging. (8)

Ans. Energy consumed in one minute with rated current and 0.8 pf lagging.

$$= 240 \times 10 \times 0.8 \times (1/60) \times 10^{-3} = 0.032 \text{ kWh}$$

$$\therefore \text{Revolutions in one minute} = 0.032 \times 600 = 19.2$$

$$\therefore \text{Speed of disc} = 19.2 \text{ rpm}$$

When lag adjustment is altered:

$$\text{Steady speed } N = KVI \sin(\Delta - \phi)$$

If the lag adjustment is correctly done $\Delta = 90^\circ$

Under this condition steady speed

$$N = KVI \sin(90^\circ - \phi) = KVI \cos \phi$$

$\therefore$ Error introduced because of incorrect lag adjustment

$$= \frac{KVI[\sin(\Delta - \phi) - \cos \phi]}{KVI \cos \phi} \times 100 = \frac{\sin(\Delta - \phi) - \cos \phi}{\cos \phi} \times 100 \text{ percent}$$

we have $\Delta = 86^\circ$

(a) at unity p.f. $\phi = 0^\circ$

$$\therefore \text{Error} = \frac{\sin(86^\circ - 0^\circ) - 1}{1} \times 100 = -0.24\%$$

(b) At 0.5 p.f. lagging, $\phi = 60^\circ$

$$\text{Error} = \frac{\sin(86^\circ - 30^\circ) - \cos 60^\circ}{\cos 60^\circ} \times 100$$

$$= -12.3\%$$

Q.3. (b) Explain the working of electro resonance type power factor meter. Draw the phasor diagram under different power factor conditions. (7)

Ans: It consists of a laminated iron core. On one end of the core a fixed coil is wound which is called magnetizing coil. This coil is connected across the supply whose frequency is to be measured. This coil carries current which has same frequency as that of supply.

On the same core, a moving coil is pivoted which carries a pointer. A capacitor C is connected across the terminals of the fixed coil.

Let

I = Current through magnetizing coil
 ϕ = Flux in the iron core

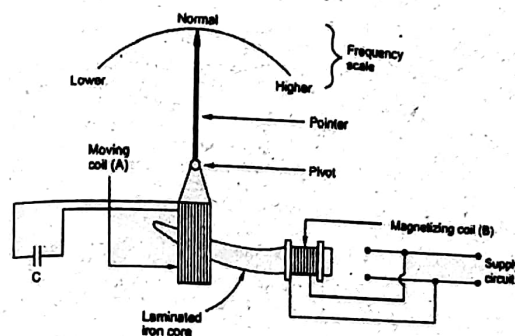


Fig. Electrical resonance type frequency meter

The flux ϕ is assumed to be in phase with the current I .

This flux induces the voltage in the moving coil which always lags flux ϕ by 90° .

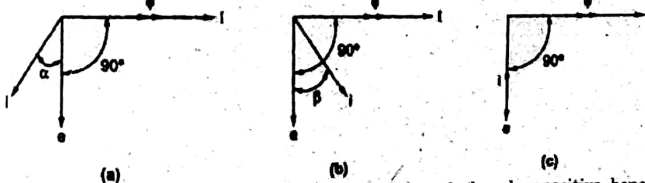
Let i = Current through moving coil

The phase of the current i depends on the inductance of the moving coil and the capacitor C .

Consider the different cases and the corresponding phasor diagrams to understand the working of the meter, as shown in the Fig. 2. (a), (b) and (c).

In Fig. 2 (a) the circuit of moving coil A is assumed to be inductive, hence current i lags the induced voltage e by angle α . Hence the torque acting on the moving coil is given by,

$$T_d \propto Ii \cos(90^\circ + \alpha)$$



In Fig. 2 (b) the circuit of moving coil A is assumed to be largely capacitive, hence current i leads the induced voltage e by angle β . Hence the torque acting on the moving coil is given by,

$$T_d \propto Ii \cos(90^\circ - \beta)$$

This torque is in opposite direction to the torque produced in case of inductive nature of the moving coil circuit.

The Fig. 2 (c) shown the resonance condition where the inductive reactance is equal to the capacitive reactance. So current i is in phase with e and the torque acting on the moving coil is given by,

$$T_d \propto I \cos(90^\circ) = 0$$

Hence under resonance condition, torque acting on the moving coil is zero.

Now the capacitive reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ is constant for a given frequency. But the inductive reactance $X_L = \omega L$ not only depends on the frequency but also depends on the position of the moving coil on the core. Nearer the moving coil to the magnetizing coil, higher is its inductance. Thus for a given frequency, moving coil moves in such a way to achieve a position where $X_L = X_C$ and electrical resonance is achieved. At this position, torque on the moving coil is zero and the pointer indicates the corresponding frequency. The design of the instrument is such that for a normal frequency, the coil takes a mean position. The capacitor C is chosen such that electrical resonance takes place at this mean position and pointer indicates the normal frequency.

If frequency is higher than the normal value, then $X_C = \frac{1}{2\pi f C}$ decreases. Hence $X_L = 2\pi f L$ must decrease in order to achieve resonance. So moving coil moves away from the magnetizing coil on the core and pointer moves to the right of the mean position, indicating higher frequency.

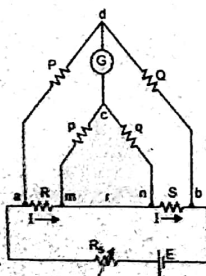
If frequency is lower than normal value, $X_C = \frac{1}{2\pi f C}$ increases. So to achieve $X_L = X_C$, the moving coil moves towards the magnetizing coil where inductance increases. Thus pointer moves to the left of the mean position, indicating the lower frequency.

An important advantage of the instrument is that the great sensitivity is achieved as the inductance of the moving coil changes slowly with variation of its position on the core. This meter is also called ferrodynamoc frequency meter.

UNIT-II

Q.4 (a) Draw the circuit of Kelvin double bridge used for measurement of low resistance. Derive the condition for balance. (8)

Ans. Kelvin Double Bridge: The kelvin double bridge incorporates the idea of a second set of ratio arms - hence the name double bridge- and the use of four terminal resistors for the low resistance arms. Figure shows the schematic diagram of Kelvin double bridge. The first ratio arm is P and Q . The second set of ratio arms p and q is used to connect the galvanometer to a point d at the appropriate potential between points m and n to eliminate the effect of connecting lead resistance r between the unknown resistance R and the standard resistance S .



The ratio p/q is made equal to P/Q . Under balance conditions there is no current through the galvanometer which means that the voltage drop between a and b , E_{ab} is equal to voltage drops E_{amd} between a and c .

$$E_{ab} = \frac{P}{P+Q} E_{ac} \text{ and } E_{ac} = I \left[R + S + \frac{(p+q)r}{p+q+r} \right]$$

$$\text{and } E_{amd} = I \left[R + \frac{p}{p+q} \left(\frac{(p+q)r}{p+q+r} \right) \right] = I \left[R + \frac{pr}{p+q+r} \right]$$

for zero galvanometer deflection, $E_{ab} = E_{amd}$

$$\frac{PI}{P+Q} \left[R + S + \frac{(p+q)r}{p+q+r} \right] = I \left[R + \frac{pr}{p+q+r} \right]$$

$$\text{or } R = \frac{P}{Q} S + \frac{qr}{p+q+r} \left[\frac{P}{Q} - \frac{p}{q} \right] \text{-----(1)}$$

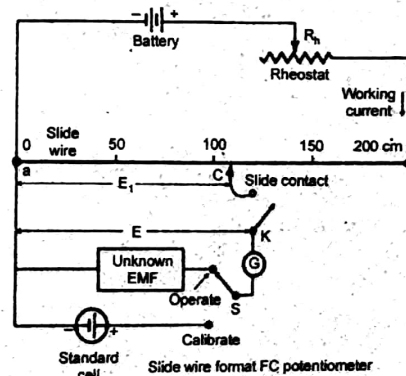
$$\text{not if } \frac{P}{Q} = \frac{p}{q} \text{ Eq(1) becomes, } R = \frac{P}{Q} S \text{-----(2)}$$

Eq (2) is the usual working equation for the kelvin bridge. It indicates that the resistance of connecting lead, r , has no effect on the measurement, provided that the two sets of ratio arms have equal ratios.

Q.4. (b) Describe the basic principle of operation of a D.C. potentiometer. (7)

Ans. Basic Principle of operation of a DC Potentiometer

With switch 'S' in the "operate" position and the galvanometer key K open, the battery supplies the "working current" through the rheostat R_h and the slide wire. The working current through the slide wire may be varied by changing the rheostat setting.



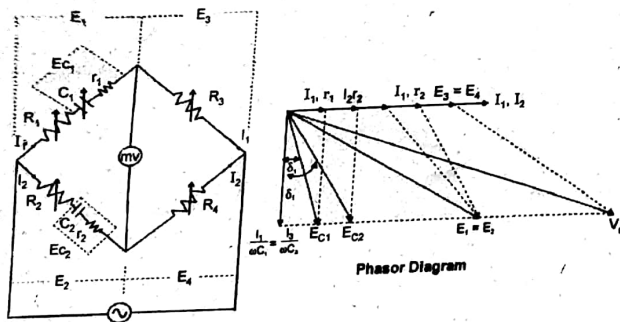
Circuit Diagram of a basic slide wire potentiometer

The method of measuring the unknown voltage E , depends upon finding a position for the sliding contact such the galvanometer shows zero deflection, i.e., indicates null condition, when the galvanometer key, K , is closed. Zero galvanometer deflection or a null means that the unknown voltage, E is equal to the voltage drop E_1 , across portion ac of the slide wire. This determination of the value of unknown voltage now becomes a matter of evaluating the voltage drop E_1 along the portion ac of the slide wire.

The slide wire has a uniform cross-section and hence uniform resistance along its entire length. A calibrated scale in cm and fractions of cm. is placed along the slide wire so that the sliding contact can be placed accurately at any desired position along the slide wire. Since the resistance of slide wire is known accurately, the voltage drop along the slide wire can be controlled by adjusting the value of working current. The process of adjusting the working current so as to match the voltage drop across a portion of sliding wire against a standard reference source is known as "Standardisation".

Q.3 (a) Describe how an unknown capacitance can be measured with the help of D'Sauty bridge. What are the limitations? (8)

Ans. D'Sauty Bridge measures an unknown capacitance in term of a standard capacitance i.e. comparing two capacitance. The connection diagram is shown below.



D'Sauty Bridge

- C_1 = A standard capacitor.
- C_2 = A standard capacitor whose capacitance has to be measured.
- r_1, r_2 = Representing loss of their corresponding capacitors.
- R_1, R_2 = Non-inductive resistance.
- R_3, R_4 = They are connected in series with C_1 and C_2 respectively.

At balance condition

$$\left(R_1 + r_1 + \frac{1}{j\omega C_1} \right) \cdot R_4 = \left(R_2 + r_2 + \frac{1}{j\omega C_2} \right) \cdot R_3$$

$$(r_1 + R_1) \cdot R_4 + \frac{R_4}{j\omega C_1} = (R_2 + r_2) \cdot R_3 + \frac{R_3}{j\omega C_2}$$

equating both the real and imaginary parts and separate them,

$$\frac{R_4}{R_3} = \frac{(R_1 + r_1)}{(R_2 + r_2)}$$

$$D_1 = \tan \delta_1 - \omega C_1 r_1 \quad \text{and} \quad D_2 = \tan \delta_2 - \omega C_2 r_2$$

From the above equation

$$\frac{C_2}{C_1} = \frac{(R_1 + r_1)}{(R_2 + r_2)}$$

$$C_2 = \frac{R_3}{R_4} C_1$$

$$C_2 = \frac{(R_1 + r_1)}{(R_2 + r_2)} C_1$$

$$C_2 R_2 + C_2 r_2 = C_1 R_1 + C_1 r_1$$

$$\text{Multiplying both side by } \omega, \quad \omega C_2 R_2 + \omega C_2 r_2 = \omega C_1 R_1 + \omega C_1 r_1$$

$$\frac{C_2}{C_1} = \frac{(R_1 + r_1)}{(R_2 + r_2)} = \frac{R_4}{R_3}$$

$$D_2 - D_1 = \omega C_2 \left(\frac{R_1 R_4}{R_3} - R_2 \right)$$

The balance may be obtained by variation of resistances R_1, R_2, R_3 and R_4 .

The angles δ_1 and δ_2 are the phase angles of capacitor C_1 and C_2 . The dissipation factor for corresponding capacitor are

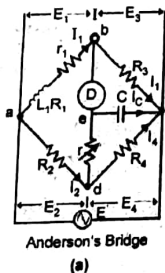
Therefore, if the dissipation factor of one of the capacitors is known the dissipation factor for the other can be determined.

Limitations: This method does not give accurate results for dissipation factor since its value depends on difference of quantities $R_1 R_4 / R_3$ and R_2 . These quantities are moderately large and their difference is very small and since this difference cannot be known with

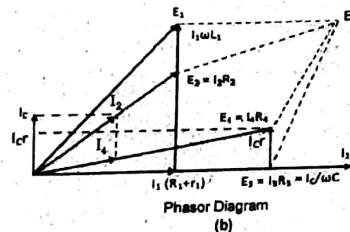
a high degree accuracy, the dissipation factor cannot be determined accurately.

Q.5 (b) Derive the equation of balance for an Anderson bridge. Discuss the advantages and disadvantages of the bridge. (7)

Ans. Anderson's Bridge: This bridge is a modification of the Maxwell's inductive and capacitive bridge. In this method, the self inductance is measured in terms of a standard capacitor. This method is applicable for precise measurement of self inductance over wide range of values. Figure 1 shows the circuit diagram of the bridge for balance conditions.



Anderson's Bridge (a)



Phasor Diagram (b)

let

- L_1 = self inductance to be measured
- R_1 = resistance of self inductor,
- r_1 = resistance connected series with self inductor

r_1, R_2, R_3, R_4 = known non-inductive resistances
 C = Fixed standard capacitor.

At balance Condition,

$$I_1 = I_3 \text{ and } I_2 = I_4 + I_4$$

Now,

$$I_1 R_3 = I_C \times \frac{1}{j\omega C}$$

$$I_C = I_1 \omega C R_3$$

The other balance condition equation is expressed as

$$I_1(r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_C r$$

$$I_C \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_C) R_4$$

By substituting the value of I_C in the above equation we get,

$$I_1(r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_1 j\omega C R_3 r$$

$$I_1(r_1 + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2$$

and $I_1(R_3 + j\omega R_3 R_4 + j\omega C R_3 r) = I_2 R_4$
on equating the equation, we get

$$I_1(r_1 + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2$$

$$= I_1 \left(\frac{R_1 R_2}{R_3} + \frac{j\omega C R_3 r R_2}{R_4} + j\omega C R_3 R_2 \right)$$

Equating the real and the imaginary part, we get

$$R_1 = \frac{R_1 R_2}{R_4} - r_1$$

$$L_1 = C \frac{R_3}{R_4} [4(R_4 + R_2) + R_2 R_4]$$

Advantages of Anderson's Bridge:

- (1) In Anderson's bridge, it is very easy to obtain the balance point as compared to max well's bridge.
- (2) In this bridge, a fixed standard capacitor is used therefore there is no need of costly variable capacitor.
- (3) This method is very accurate for measurement of capacitance in terms of inductance.

Disadvantages of Anderson's Bridge:

- (1) It is more complex as compared with Maxwell's inductance bridge. It has more parts and hence complex in setting up and manipulate.
- The balance equations of Anderson's bridge are quite complex and much more tedious.
- (2) An additional junction point increases the difficulty of shielding the bridge.

UNIT-III

Q.6 (a) What is XY recorder? Give the application and draw the circuit diagram. (7)

Ans. X-Y Recorder: X-Y recorder is an instrument which gives a graphic record of the relationship between two variables in X-Y recorder, an emf is plotted as a function

of another emf. This is done by having self-balancing potentiometer control the position of the rolls while another self-balancing potentiometer controls then position of the recording pen. An XY recorder consists of a pair servo-systems, driving a recording pen in two axes through a proper sliding pen and moving arm arrangement, with reference to a stationary paper chart.

Fig. 1 shows a block (or circuit) diagram of a typical XY recorder. A signal enters each of the two channels. The signal are attenuated to the inherent full scale range of the recorder, the signal then passes to a balance circuit where it is compared with an internal reference voltage.

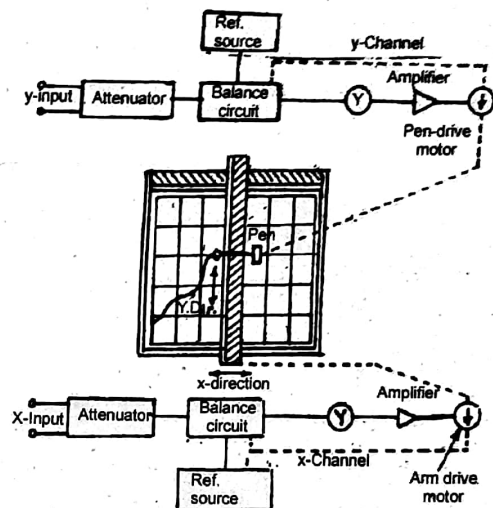


Fig. X-Y Recorder

Applications

The use of XY recorders in laboratories greatly simplifies and expedites many measurements and tests. A few examples are

1. Speed torque characteristics of motors.
2. Lift Drag wind tunnel tests.
3. Plotting of characteristics of vacuum tubes, zener diodes rectifiers and transistors, etc.
4. Regulation curves of power supplies.
5. Plotting stress-strain curves, hysteresis curves and vibrations amplitude against swept frequency.
6. Electrical characteristics of materials such as resistance versus and temperature plotting the output from electronic calculators and computers.

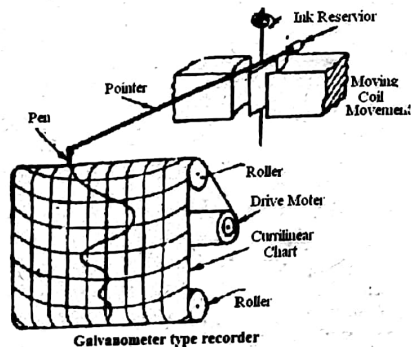
Q.7. Discuss the working of galvanometric type strip chart recorder with the help of diagram.

Ans. Principle of operation: This type of recorder operates on the deflection principle. The deflection is produced by a galvanometer which produced a torque on the coil of the recorder. This current is proportional to the quantity being measured. A galvanometer type recorder is shown in fig. the pointer is equipped with a recording mechanism (stylus). As the current flows through the coil, it deflects.

The greater the amplitude of the incoming signal, the greater is the deflection. The chart is driven at a constant speed by a clockwork mechanism or an electric motor.

When the pointer comes to rest on account of controlling torque exerted by springs, the stylus also comes to rest. Thus the value of the physical quantity is recorded.

The recorders can work on ranges from a few mA to several mA or from a few mV to several mV. They have a narrow bandwidth of 0 to 10 Hz. If it has a sensitivity of 0.4 mV/mm or from a chart of 100 mm width, a full scale deflection of 40 mV is obtained.



Q.7. Describe the construction and working of an analog storage oscilloscope. Discuss the principle of secondary emission. (15)

Ans. Analog Storage Oscilloscopes: Storage oscilloscopes, capable of high persistence phosphors, have many applications. Example of these is the capture and storage of transients and the steady display of a very low frequency signal. Two techniques are used to store signals in an oscilloscope and these are called analog and digital storage.

Analog storage is capable of higher speeds, but is less versatile than digital storage.

Principle of Secondary Emission: Fig. 1 shows a simple electrode arrangement to illustrate this principle. When a beam of electrons from the electron gun strikes the target it emits secondary electrons, which are gathered by the collector. The collector is at positive potential of the target is at a potential of which can be varied and the electron gun is at ground potential.

The value of the current in the primary electron beam, coming from the electron gun and is the electron current emitted from the target and collected by the collector, by the collector, the ratio is called the **secondary emission ratio**. The value of this ratio depends on the primary electron velocity and intensity and on the chemical composition of the target.

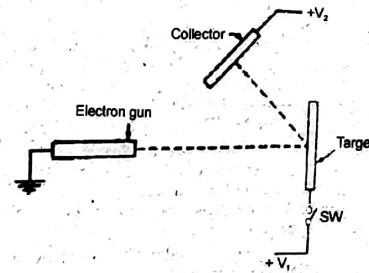


Fig. 1. Simple electrode arrangement to illustrate secondary emission

Fig. 2 will show the variation of the secondary emission ratio with target voltage, for the arrangement of when the target voltage is much greater than the collector voltage all secondary electrons emitted from the target are attracted back to it. Therefore the collector current or secondary current is zero and so also is the secondary emission ratio. The operating point is now well to the right of V_2 .

If the target voltage is slightly negative, as at point f, then all the electrons from the gun are deflected on to the collector before they reach the target.

Therefore although there is no secondary emission the collector current equals the beam current and the secondary emission ratio is unity. This point is known as the lower stable point. As the target voltage increases from these point electrons are attracted away from the collector, but they do not have enough energy to release secondary electrons from the largest.

Therefore the secondary emission ratio falls to a minimum at A. Beyond the minimum point secondary emission from the target starts to occur and these electrons are accumulated by the collector, so increasing the secondary emission ratio.

The secondary emission ratio increase though the crossover point E until it reaches a peak at C. Beyond this point secondary electron emitted from the target are attracted back in numbers greater than those which reach the collector, so that the secondary emission ratio decreases sharply. The curve reaches the upper stable point at G, where the primary and secondary currents are equal and then decreases to zero. The lower and upper stable points represent the erased and written conditions of the CRT screen and in the absence of a target voltage the can remain in one of these two stages only.

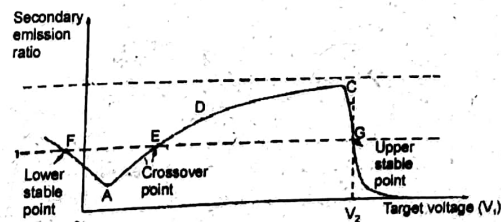


Fig. 2. Variation of Secondary emission ratio for the electrode arrangement shown in fig.

Variable Persistence Storage: It is an analog storage oscilloscope is also known as halftone storage or mesh storage.

Construction: Given in Fig. 3 illustrates the construction of a CRT using this storage technique. There are two screens, a storage mesh which retains the image traced on it by the writing gun and the phosphor screen, which is very conventional CRT. Dielectric material, consisting of a thin layer of material such as magnesium fluoride, is deposited on the storage mesh and this acts as the storage target.

The writing gun is at a high negative voltage, the float given at a few volts, negative the collector mesh at about 100 V positive and the storage mesh at ground potential or a few volts negative. The collimator consists of a conductive coating on the inside surface of the CRT. It is biased so as to distribute the flood gun electrons evenly over the surface of the target and causes the electron to be perpendicular to the storage mesh.

Working: When the writing gun is aimed at the storage target it causes areas where it strikes to be charged to a positive potential, due to secondary emission. These areas are maintained at their upper stable point, even after the writing gun is switched off, due to the action of the flood gun. Electrons from the flood gun also pass through. Those areas which are positively charged causing the phosphor beyond to glow and displaying the original signal traced by the writing gun.

The pattern stored on the storage target lasts for about one hour, but it can display a bright image for about one minute. The stored pattern fades due to electrons from the flood gun charging other parts of the storage surface, giving an impression that the whole screen has been written. This is known as fading positive. To erase an image which has been stored the storage mesh is momentarily raised to the same positive potential as the collector mesh.

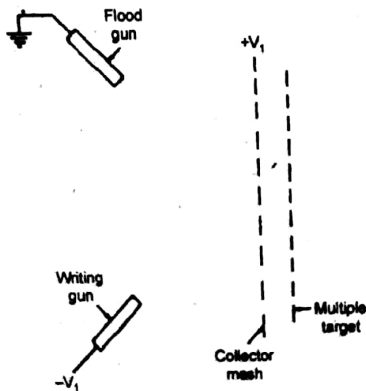


Fig. 3. Multiple target and flood gun arrangement.

Application: Such as the storage of an entire waveform of a slow moving signal, which then fades the next trace is written.

It can also be used to store several traces before the first one fades, so as to see how the signal changes with time.

UNIT-IV

Q.8. Explain the construction and working of an average reading VTVM. Describe the advantages and disadvantages. How this meter can be converted to a rms reading meter. (15)

Ans. Construction and Working of average reading VTVM
Average reading Diode vacuum Tube voltmeters.

The simplest type of diode voltmeter is similar to a rectifier instrument. fig. 1(a) show a schematic diagram for an average reading voltmeter using a vacuum tube diode. The arrangement include a vacuum tube diode, a resistance R, and a PMMC instrument all connected in series. The applied voltage is $e = E_m \sin \omega t$.

A high series resistance R is desirable both to limit the current and to make the plate voltage-current characteristics linear. A linear plate characteristics are essential in order to make the current directly proportional to voltage. The plate resistance of diode is not constant and hence in order to swamp out the variations in plate resistance, a high value of series resistance, R is used. If the series resistance is sufficiently high the resistance of the tube is negligible in comparison, and therefore we get a linear plate voltage current characteristics there by the scale of PMMC instrument is uniform and is independent of the variations of tube resistance. A series resistance of $10^5 \Omega$ or more is used of the purpose of measurement of a sinusoidal input voltage as shown in fig. 1. (a). The diode conduct during the positive half cycle and does not conduct during the negative half cycle. The average current through the meter is thus

$$I_{av} = \frac{E_{av}}{2R} = \frac{E_{rms}}{2 \times 1.11 \times R} = 0.45 \frac{E_{rms}}{R} \quad (1)$$

where E_{rms} and E_{av} are respectively the rms and average values of applied voltage. The factor 2 is introduced because the voltmeter works on half wave rectification.

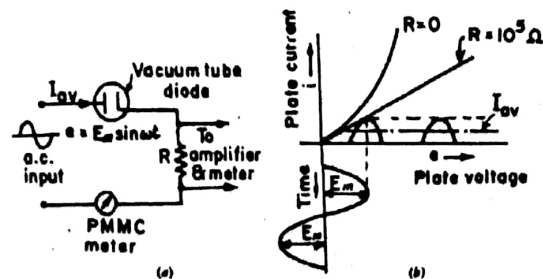


Fig. 1. Average reading vacuum tube diode voltmeter

Advantage of average reading VTVM

1. **Simplicity,** The average reading diode type meter is simplest in construction.
2. **High input resistance and low power consumption.** The input resistance of the circuit depends upon the value of resistance, R which ranges from $10^5 \Omega$ to $1 \text{ M}\Omega$ and its resistance is higher than that of multimeter circuits for a.c. voltage measurements. In most diode voltmeters, indication is obtained from the output of d.c. amplifier, the output of which is connected across the resistance R. This makes it possible to replace a relatively sensitive meter by a less sensitive and rugged meter. Also the input

Resistance R can be made greatly large, of the order of tens to hundreds of $M\Omega$. This way the power consumed by the voltmeter can be made negligibly small. (This is because the power consumed by the diode circuit is $E_{rms}^2/2R$ watt).

3. Conversion to rms reading voltmeter: The rms value of voltage is usually of principle interest. The average reading voltmeter can be converted into an rms reading voltmeter by calibrating the meter scale in terms of the rms value of a sine wave. Since so many waveforms in electronics are sinusoidal, the average reading voltmeter with an rms calibrated scale is an entirely satisfactory instrument. In fact an average reading voltmeter with a calibrated rms scale is much less expensive than a true rms reading voltmeter.

Disadvantage of average reading VTVM:

1. Frequency range: The frequency range of this instrument extends well over the audio frequency range but at radio frequencies the value capacitance and the distributed capacitance of the series resistance R , upset the voltage division and make the instrument useless.

2. Response to low voltages: The instrument give a deflection proportional to the mean value during the conducting half cycle provided it is not used for voltages comparable with the voltage drop across the valve. The instrument has an evenly divided scale because of linear characteristics for higher voltages but exhibits non-linearity for lower voltages and therefore for lower voltages, a graphical treatment may have to be used to determine the voltage.

Conversion of average reading VTVM to rms reading voltmeter.

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The rms value of a voltage wave which has similar positive and Negative half cycles, is related to the average value by the form factor, for a sinusoidal voltage the form factor is 1.11. Therefore, when an average reading, voltmeter has scale making corresponding to the rms value of the applied sinusoidal input waveform, those markings are actually corrected by a factor 1.11 from the average value of applied voltage. Now sinusoidal waveform, when applied to this voltmeter, will cause it to read either high or low, depending upon the factor of the waveform. If the form factor of the applied voltage is less than that of a sinusoidal wave (i.e., less than 1.11), the meter reading is high while on the other hand if the form factor of the applied voltage is higher than 1.11, the meter reading are lower.

Q.9. Write short notes on:

(a) Digital Multimeter

Ans. Digital Multimeter

Digital multimeter (DMM) is basically a digital voltmeter and may be used for the measurement of voltage (dc or ac) and resistance. All quantities other than dc voltage are first converted into an equivalent voltage by some device. The block diagram of a basic DMM is shown in Fig. 1.

Working Operation

For measurement of ac voltage, the input voltage is converted into a dc voltage by means of a rectifier. A compensated attenuator is employed.

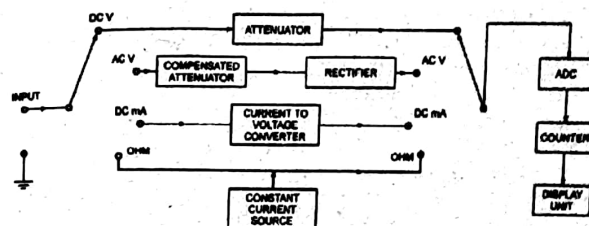


Fig. 1. Block diagram of Digital Multimeter

- For measurement of resistance, a constant current (depending on the range) supplied from a battery or a constant current source is passed through the resistance under measurement and the voltage developed across it is measured. The resistance value is displayed in ohms.
- For measurement of current, the unknown current is passed through a precision resistor in many commercial digital multimeters and the voltage developed across the precision resistor is measured. The current value is displayed in mA.
- For measurement of current, a current-to-voltage converter may also be used as shown in Fig. 2.

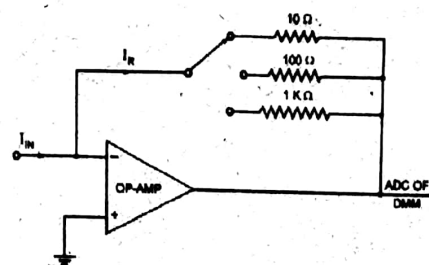


Fig. 2. Current-to-voltage converter.

Advantages of a Digital Multimeter

1. Its output display which directly shows the measured value,
2. High accuracy
3. Ability to read both positive and negative values.

Applications of Digital Multimeters

1. Temperature and Environmental Applications

- Low cost weather station
- DMM internal temperature

2. Voltage Measurements

- High and low value DC measurement
- Peak to Peak and DC average measurement

3. Current Measurements

- DC current measurement
- True RMS AC current

4. Resistance Measurement

- Micro ohm meter
- Measuring resistance with constant voltage

5. Time and Frequency measurement

- Fast frequency
- Time measurement

Q.9 (b) Triangular Wave Generator

Ans. Triangular Wave Generator

(8)

It consists of a comparator (A) and an integrator (B) as shown in figure 1. The output of comparator A is a square wave of amplitude $\pm V_{sat}$ and is applied to the inverting (-) input terminal of the integrator B. The output of integrator is a triangular wave and it is feedback as input to the comparator A through a voltage divider R_2, R_3 .

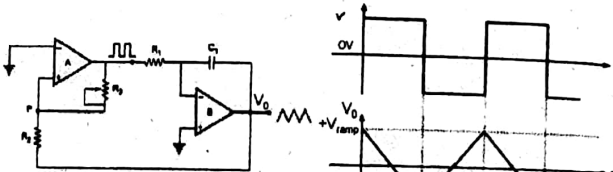


Fig. 1.

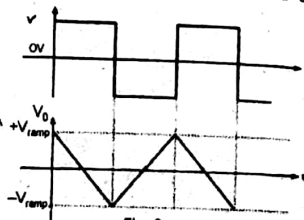


Fig. 2.

To understand circuit operation assume that the output of comparator A is at $+V_{sat}$. This forces a constant current through C_1 to give a negative going ramp at the output of the integrator as shown in the fig 2. Therefore, one end of voltage divider is at a voltage $+V_{sat}$ and the other at the negative going ramp. when the negative going ramp reaches a certain value V_{ramp} the output of comparator A switches from positive saturation to negative saturation ($-V_{sat}$). This forces a reverse constant current (right to left) through C_1 to give a positive going ramp at the output of the integrator as shown in the figure when positive going ramp reaches $+V_{ramp}$, the effective voltage at point a become slightly above 0V. As a result the output by comparator A switches from negative saturation to positive saturation ($+V_{sat}$). The sequence then repeats to give triangular wave at the output of integrator B.

210
-45
45
300

23280
1100
700
25080